

Data Communications

Lecture #1

Welcome

- This is the Data Communications course
- Today ..
 - Introduction to the course
 - Introduction to the Internet
 - Principles
 - Terminologies
 - Trend

Course Introduction

Course Introduction

- What is this course about?
 - Learn what the Internet looks like and what design principles have been applied to it
 - Discuss the latest issues and technology in the networking domain.
 - Mainly focuses on the network/link layers (c.f., Data communications has covered the application/transport layer)
- Prerequisite
 - A bit of programming skill (C / C++)

Administrative Information

- Class hour
 - 15:00 ~ 14:15 on Mon. & Thu.
- Class room
 - #411 in Hanbit Bldg.

Textbook & References

- Textbook
 - Computer Networking A Top-Down Approach, 8th Edition, James F. Kurose, Keith W. Ross, Pearson
- References
 - Online materials including technical reports, papers, tutorials will be used

Grading policy

- **Exams:** 70% (35% for midterm and final, respectively)
- **Project & Homework:** 25%
- **Class participation:** 5%

- You will **fail** this course when cheating in assignment or projects

Projects

- Team or individual projects will be provided
- For the project, you may be supposed to
 - Read a few papers and submit their summaries and critiques
 - Do a simulation (programming assignment)
 - Implement simple network application to evaluate network performance

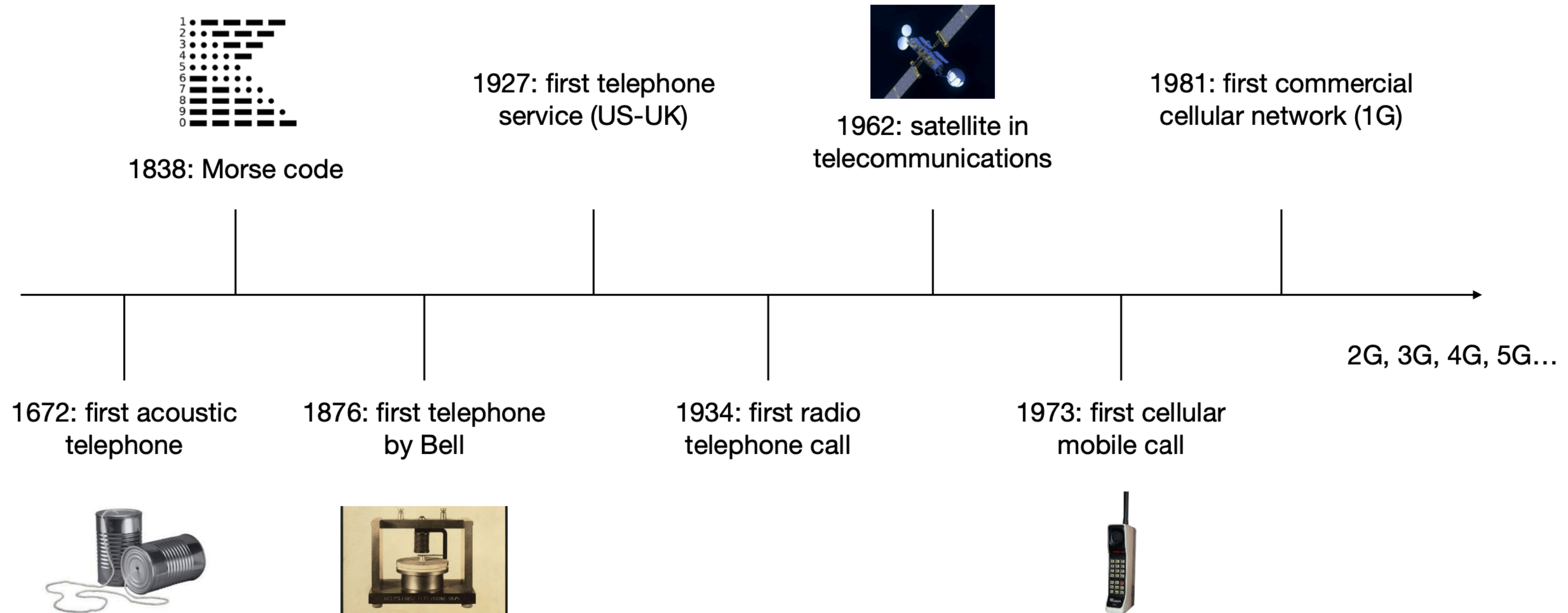
Instructor

- Dohyung Kim
 - Assistant professor in Dept. of Computer Science & Engineering
 - Leading “Intelligence in Computer Networking (ICN)” Lab
 - #403 in Hanbit bldg.
 - Email: d.kim@kangwon.ac.kr
 - <https://www.icnlab.dev>

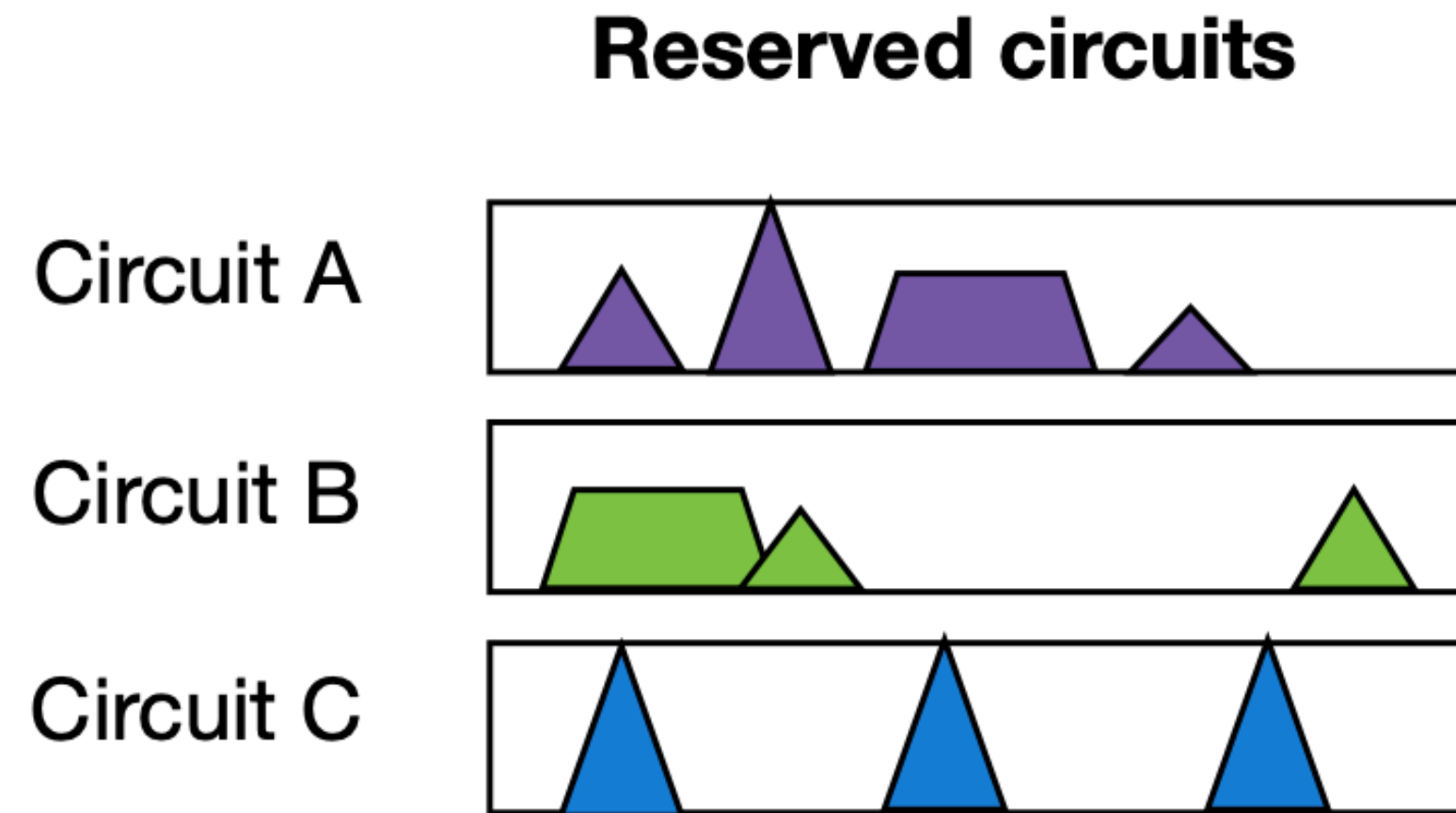
Introduction to the Internet

Telephone network vs. Internet

- Telephone network

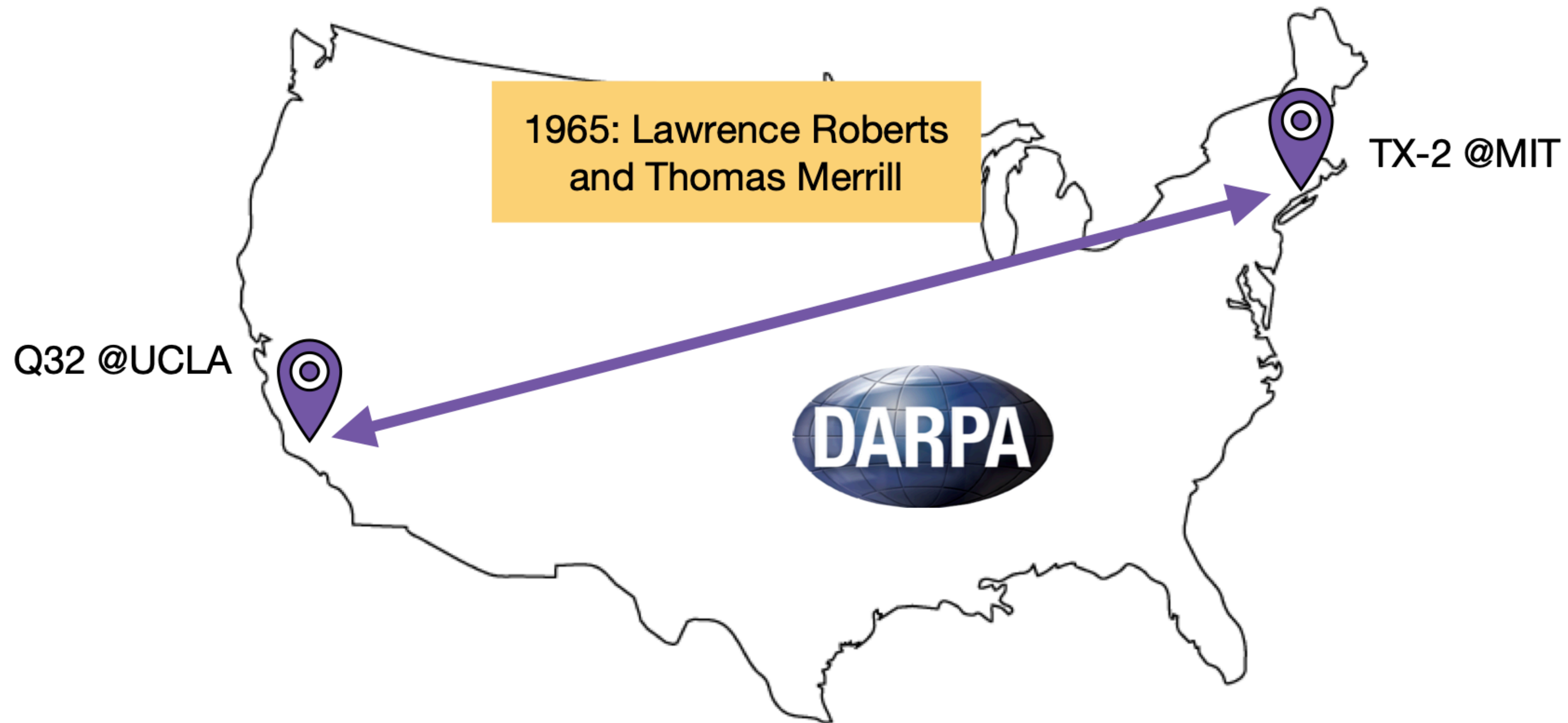


Circuit switching



- Physical channel carrying stream of data from source to destination
- Three phases: setup, data transfer, tear-down
- Data transfer involves no routing

First WAN (wide area network)

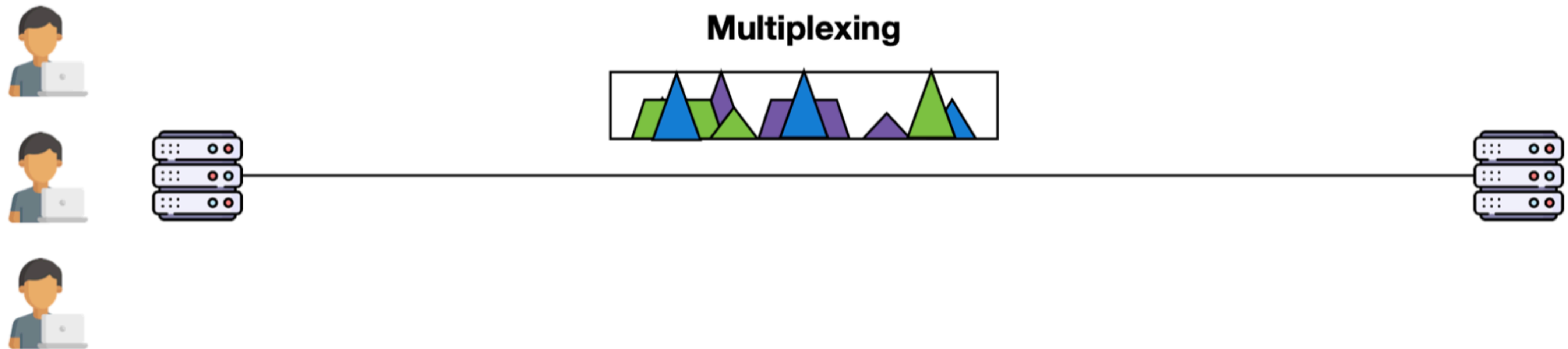


Connection is through the telephone line – it works, but it is inefficient and expensive – confirming the motivation for packet switching

Packet switching

1960s: Time-sharing
operating systems
began to emerge

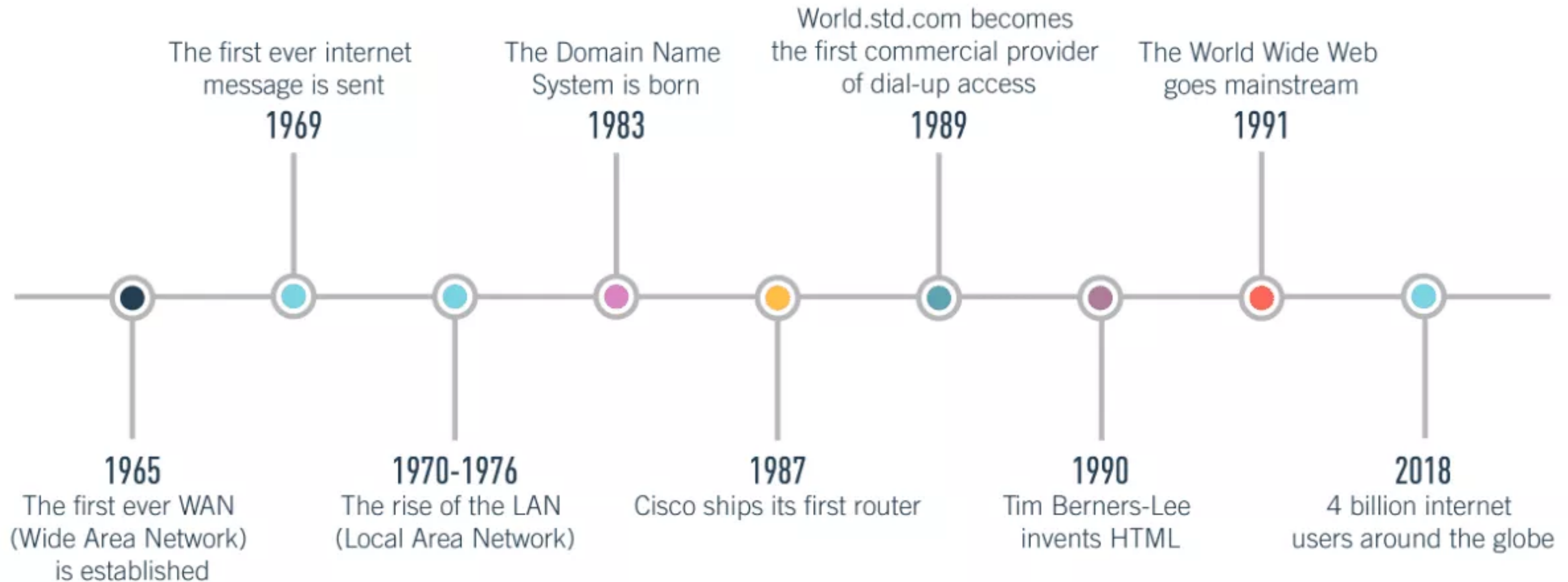
**Communication is enabled
without dedicated channels!!**



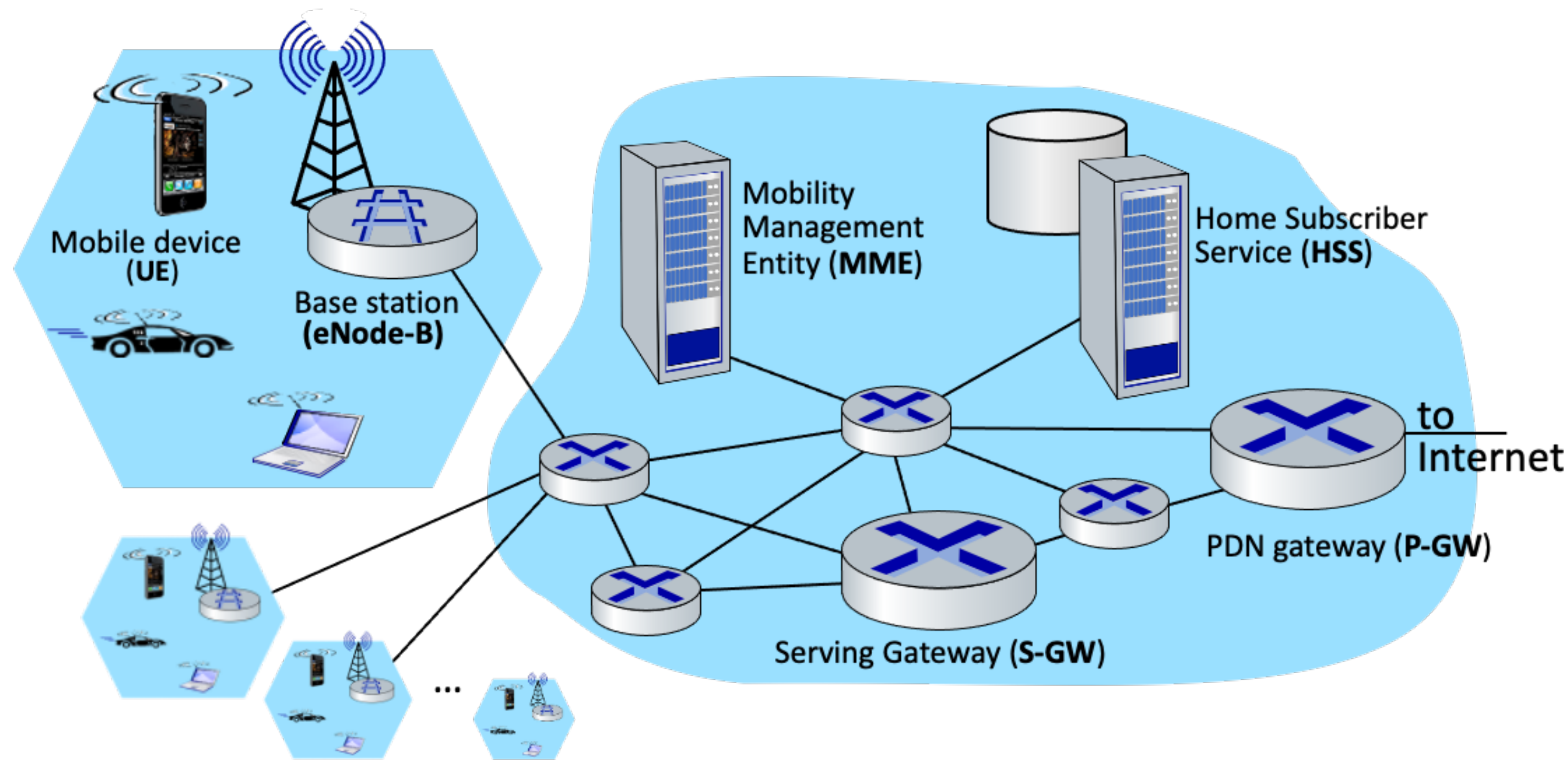
- Message broken into short packets, each handled separately
- One operation: send packet
- Packets stored/queued in each router, forwarded to appropriate neighbor

Telephone network vs. Internet

- Internet History

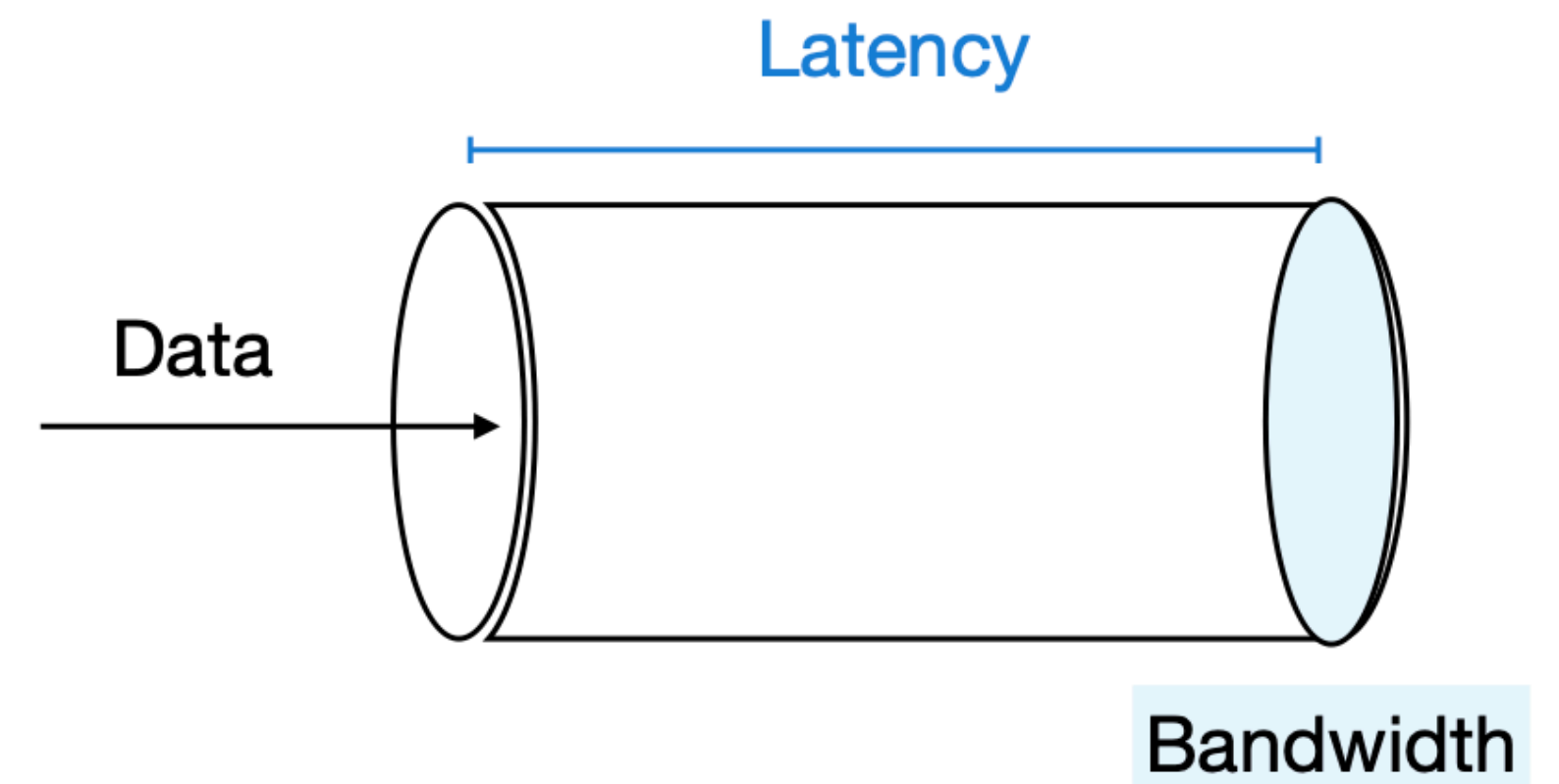


Telephone network vs. Internet

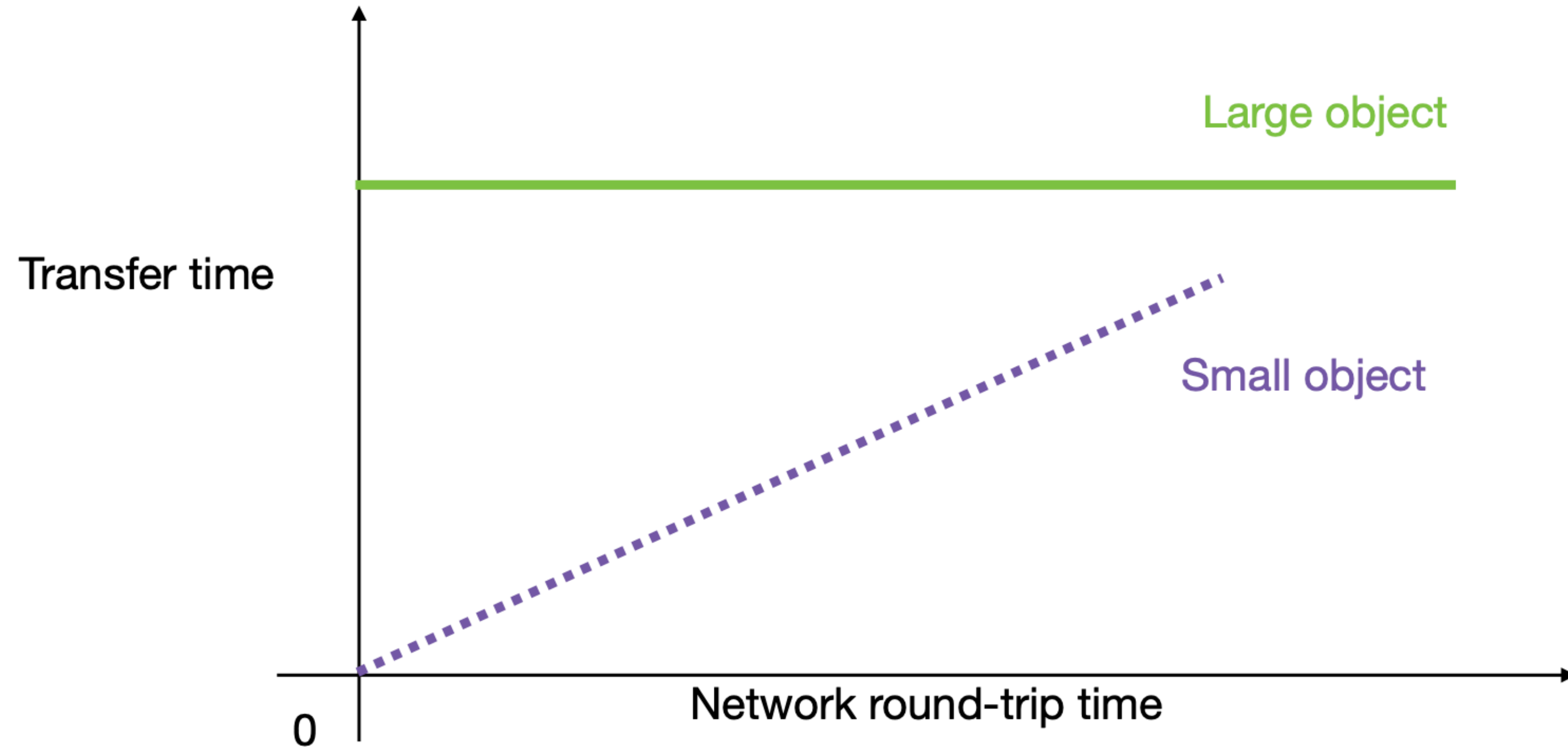


Network Performance

- Bandwidth
 - Maximum rate of data transfer across a given network path
- Latency
 - The amount of time it takes to deliver some data from the source to the destination across the network
 - “Latency lags bandwidth” - David A. Patterson



Bandwidth vs. Latency



Assume a network link with 10Gbps bandwidth and 1ms latency? How long does it take to transfer 1B, 100KB, or 10GB of data?

With **TCP**: 100KB = 1460B * (0 + 2 + 4 + 8 + 16 + 32 + 6) → 7ms in the best case, 0.0143Gbps

Performance metric

- Flow completion time
 - How long does it take to complete a traffic flow
 - How long does it take to complete a set of correlated flows (co-flows)?
- What else?
 - How long does <https://www.google.com?q=cool+network> take?
 - What is the best video quality I can watch, without the annoying “buffering”?
 - How to guarantee the per-frame latency (e.g., 20ms) in AR?
- Not just about performance
 - But also **consistent, predictable** performance

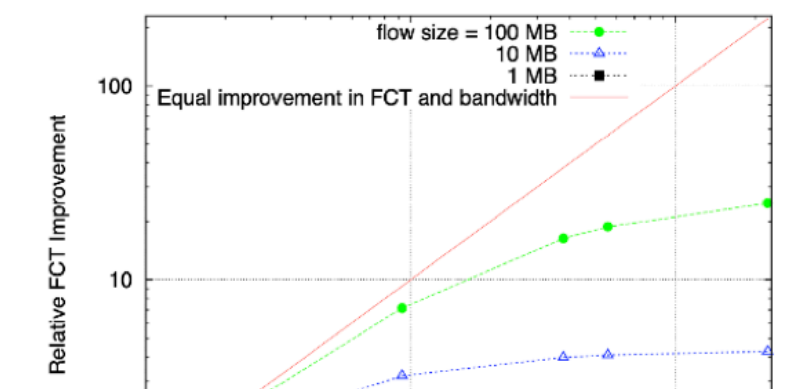
Why Flow-Completion Time is the Right Metric for Congestion Control

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ABSTRACT

Users typically want their flows to complete as quickly as possible. This makes Flow Completion Time (FCT) an important - arguably the most important - performance metric for the user. Yet research on congestion control focuses almost entirely on maximizing link throughput, utilization and fairness, which matter more to the operator than the user. In this paper we show that with typical Internet flow sizes, existing (TCP Reno) and newly proposed (XCP) congestion control algorithms make flows last much longer than necessary - often by one or two orders of magnitude. In contrast, we show how a new and practical algorithm - RCP (Rate Control Protocol) - enables flows to complete close to



ACM SIGCOMM CCR 2006

Fairness

Suppose a network is flow-fair. How useful is that?

Flow Rate Fairness: Dismantling a Religion

Bob Briscoe
BT Research & UCL
bob.briscoe@bt.com

“Both the thing being allocated (rate) and what it is allocated among (flows) are **completely daft** — both unrealistic and impractical.”

ACM SIGCOMM CCR 2007

Food for thought:

- How to translate microbenchmarks to app-level metrics?
- How to make service providers accountable?
- How to improve the performance and approach fairness?

Network Reliability

Three important considerations in network reliability

**The end-to-end
argument**

**The fate-sharing
principle**

**Packet vs. circuit
switching**

End-to-end argument

- TCP provides reliable transport
- What if no reliable transport is provided?
 - Every application that needs reliability has to engineer it from scratch: programming burden, bugs, ...
- What if the network layer tried to provide reliable delivery?

Reliable (or **unreliable**) transport

built on...

Best-effort global packet delivery

Reliable (or **unreliable**) transport

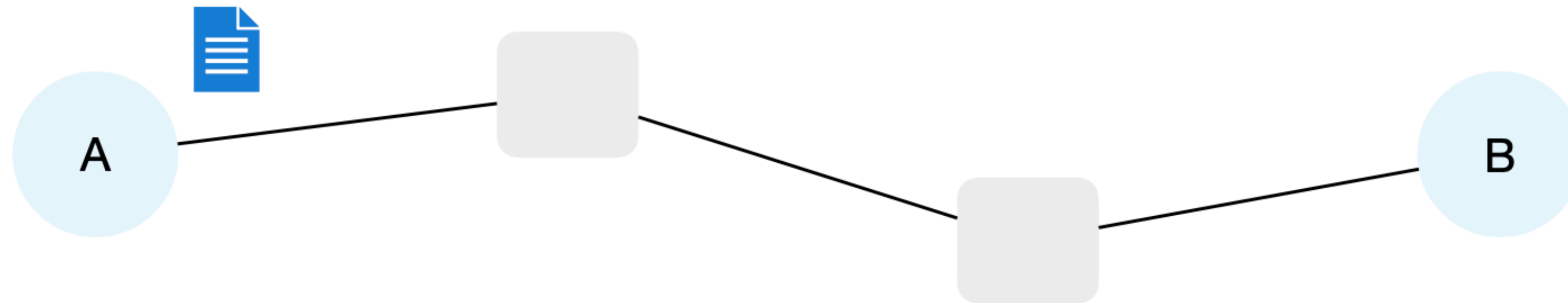
built on...

Reliable global packet delivery

What are the problems?

End-to-end argument

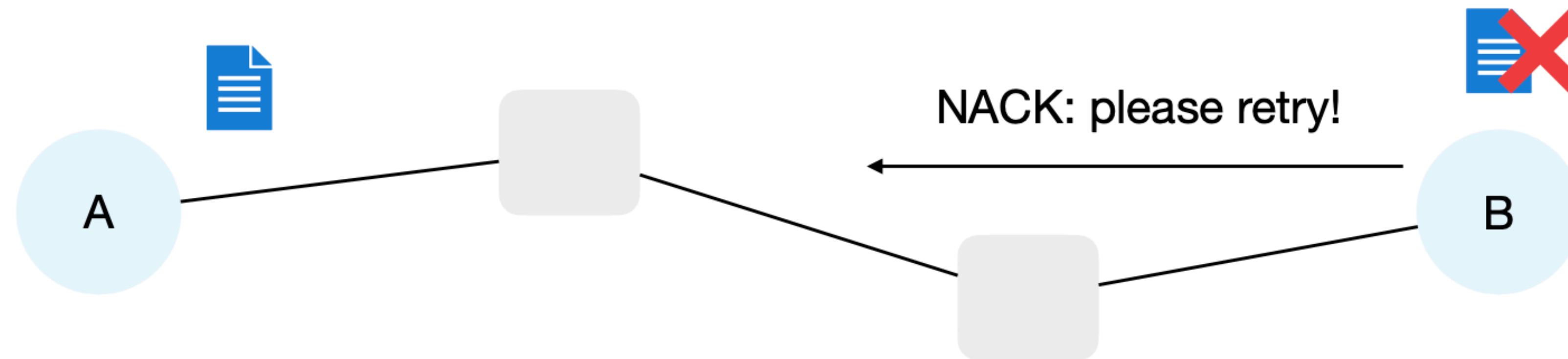
- Problem #1) there are applications that require speedy delivery, even if it's lossy
- Problem #2) by the way, can the network even achieve reliable global packet delivery?



Check reliability at every step (on the network layer).

End-to-end argument

“If a function can only be correctly implemented end-to-end, it must be implemented in the end systems. Implementing it in the network can, at best, only be an optimization.”

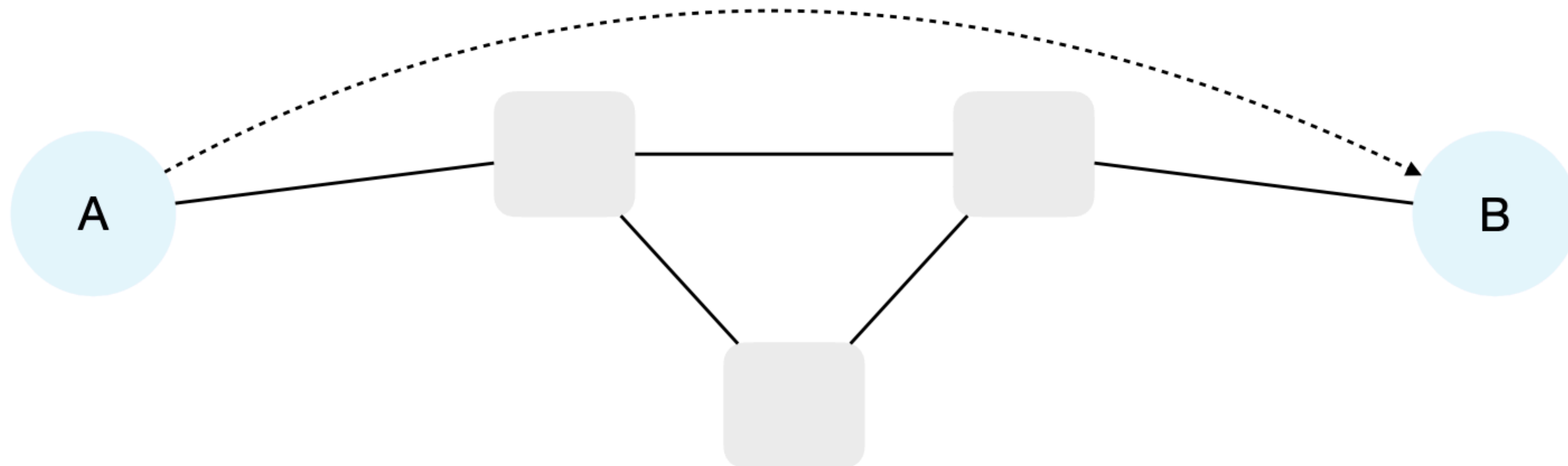


Allow unreliable steps (network layer is best-effort).
B checks correctness. On failure, B tells A to retry.

Fate-sharing principle

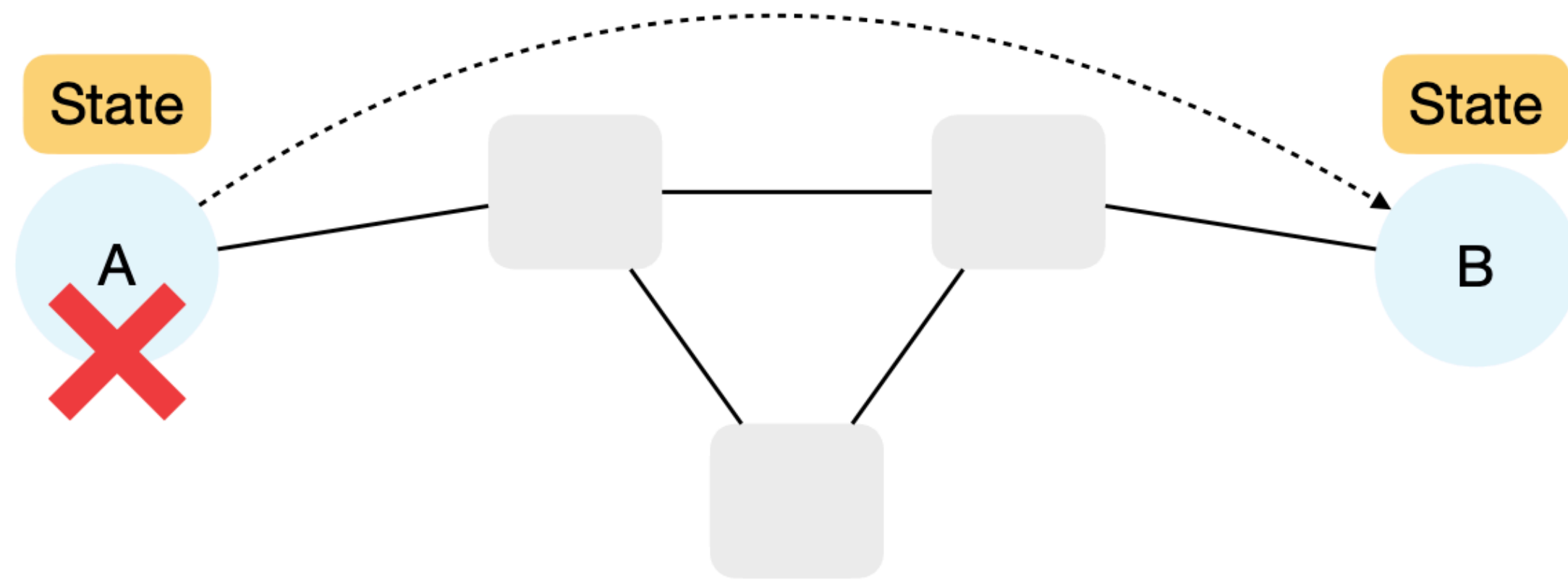
To deal with potential failures, store critical system state at the nodes which rely on that state. Only way to lose that state is if the node that relies on it fails, in which case it does not matter.

Where to maintain $A \rightarrow B$ connection state?



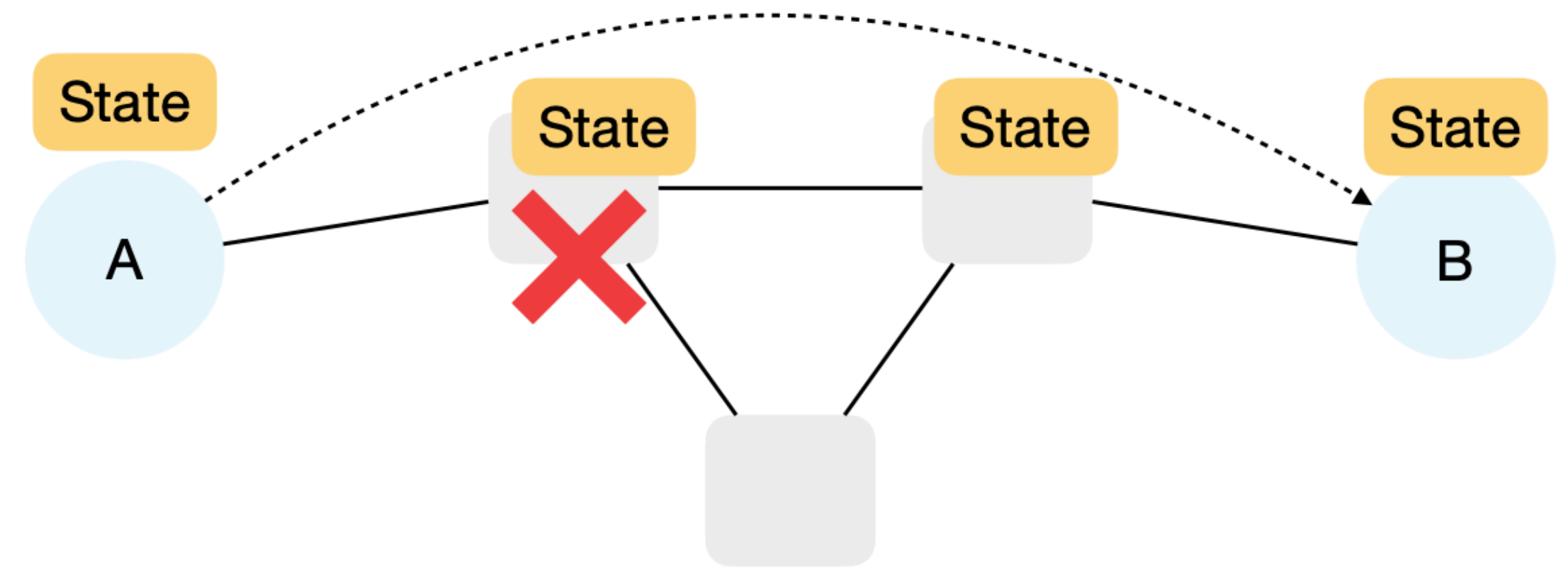
Fate-sharing principle

Keep state on end hosts



In the end-host fails, the connection state is irrelevant anyway

Keep state on network devices



Failures of network devices require the state to be cleaned up or recreated → complex consistency issue!

Packet vs. circuit switching

Circuit switching

Predictable performance

Wasteful, if packet is bursty and short

Large latency for small messages, as they wait for circuits creation

Require new circuit setup upon failure

Packet switching

Efficient use of resources

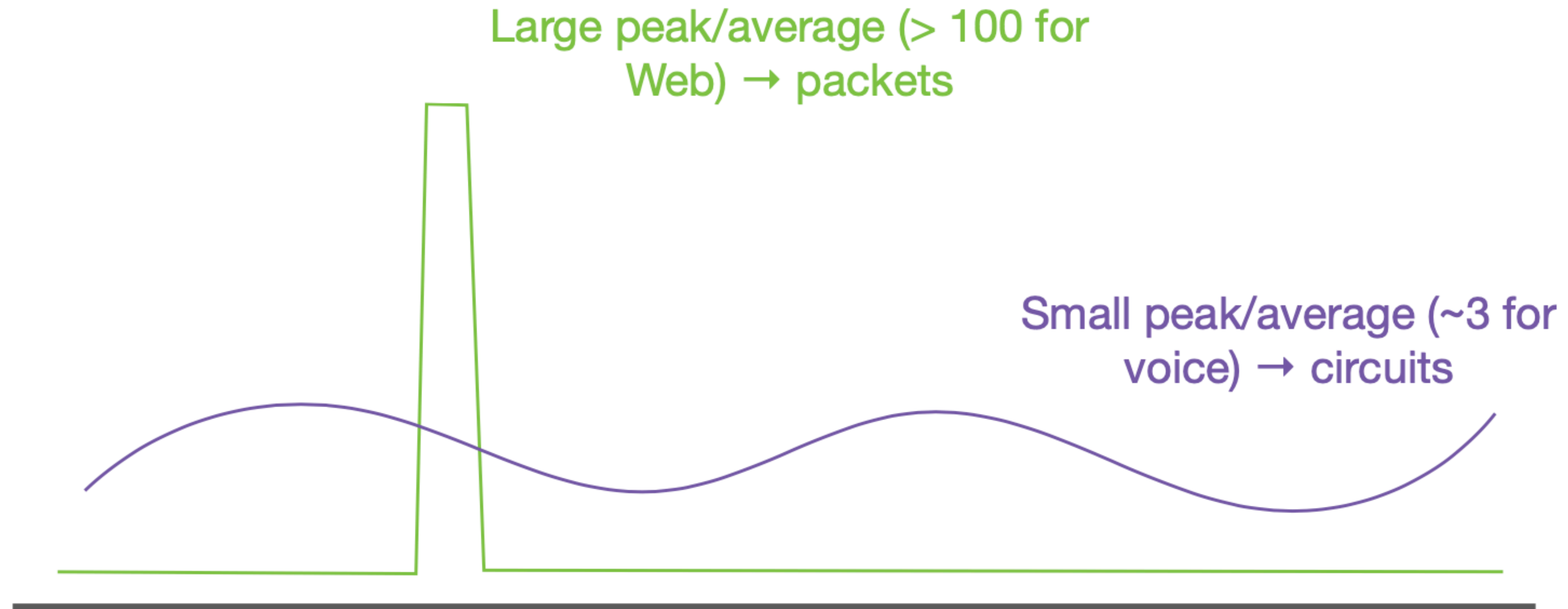
Automatic, in-network rerouting on failures

Unpredictable performance

Requires buffer management and congestion control

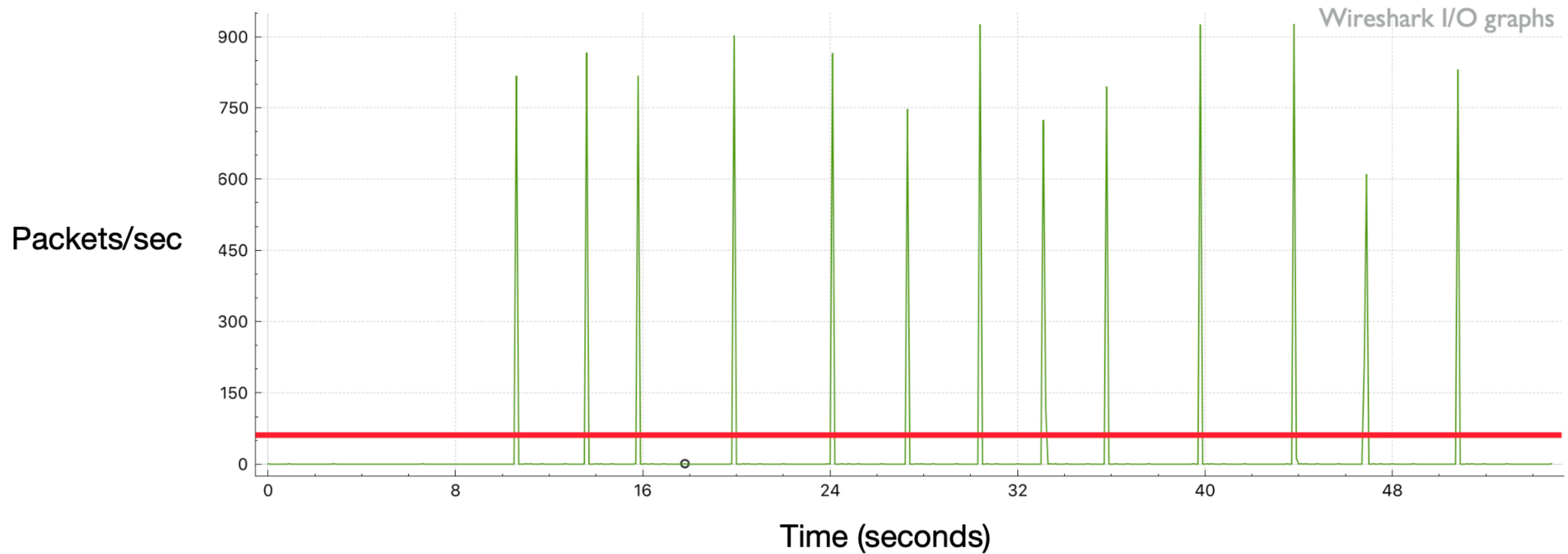
- Packet switching beats circuit switching with respect to resilience and efficiency

Packet vs. circuit switching



- Which pattern do you think modern video streaming (e.g., Youtube) will follow?

Video streaming today



- Do you know why?

Current state of Internet

- Internet is there for more than 50 years
 - Networks keep growing and more applications are developed
 - TCP/IP is the norm: to program a network application, you simply use the socket APIs
- So, the network will keep growing with the same set of technologies?
 - Partially, yes! We are still using these old technologies (TCP/IP)
 - But, there are also new developments

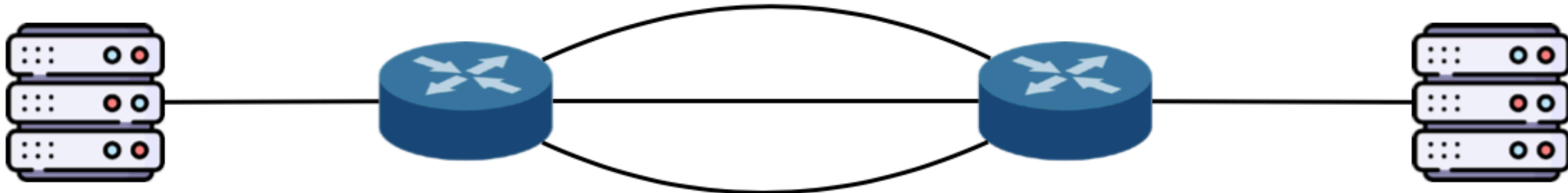
Innovation in network protocol

facebook Engineering

Open Source ▾ Platforms ▾ Infrastructure Systems ▾ Physical Infrastructure ▾ Video Engineering & AR/VR ▾

POSTED ON OCT 21, 2020 TO [ANDROID](#), [DATA INFRASTRUCTURE](#), [IOS](#), [NETWORKING & TRAFFIC](#), [WEB](#)

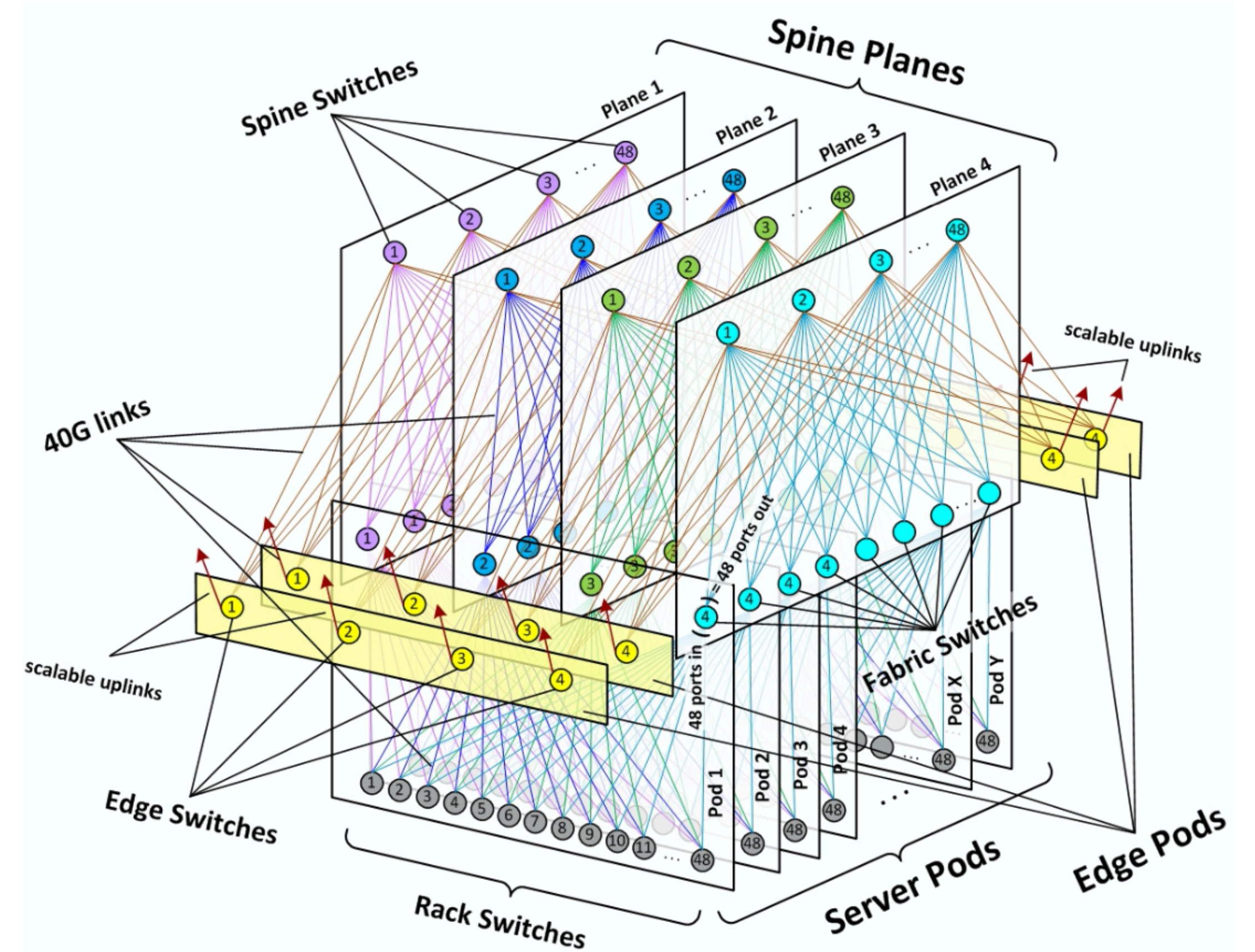
How Facebook is bringing QUIC to billions



Multipath TCP, QUIC

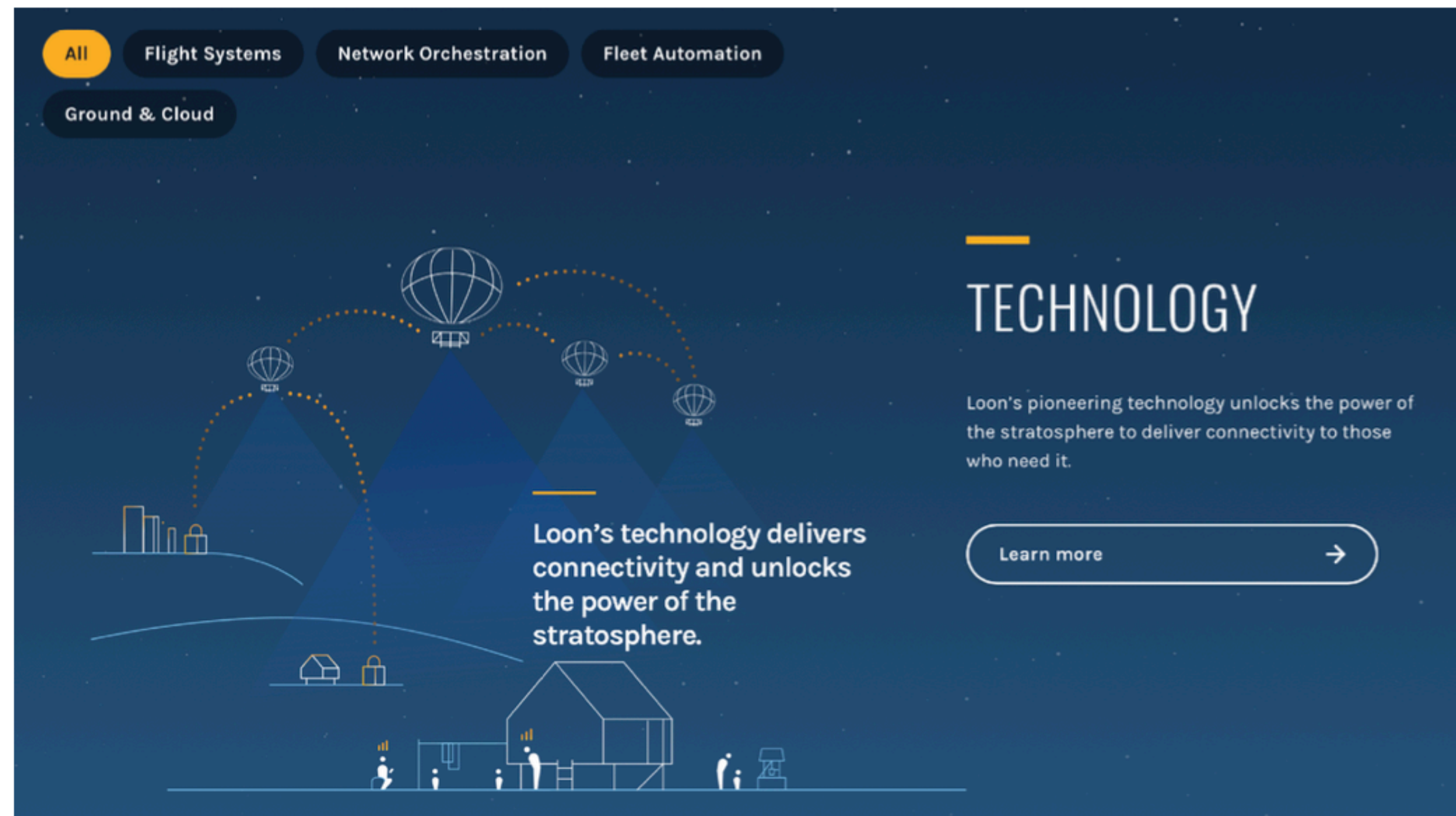


Innovations in network design

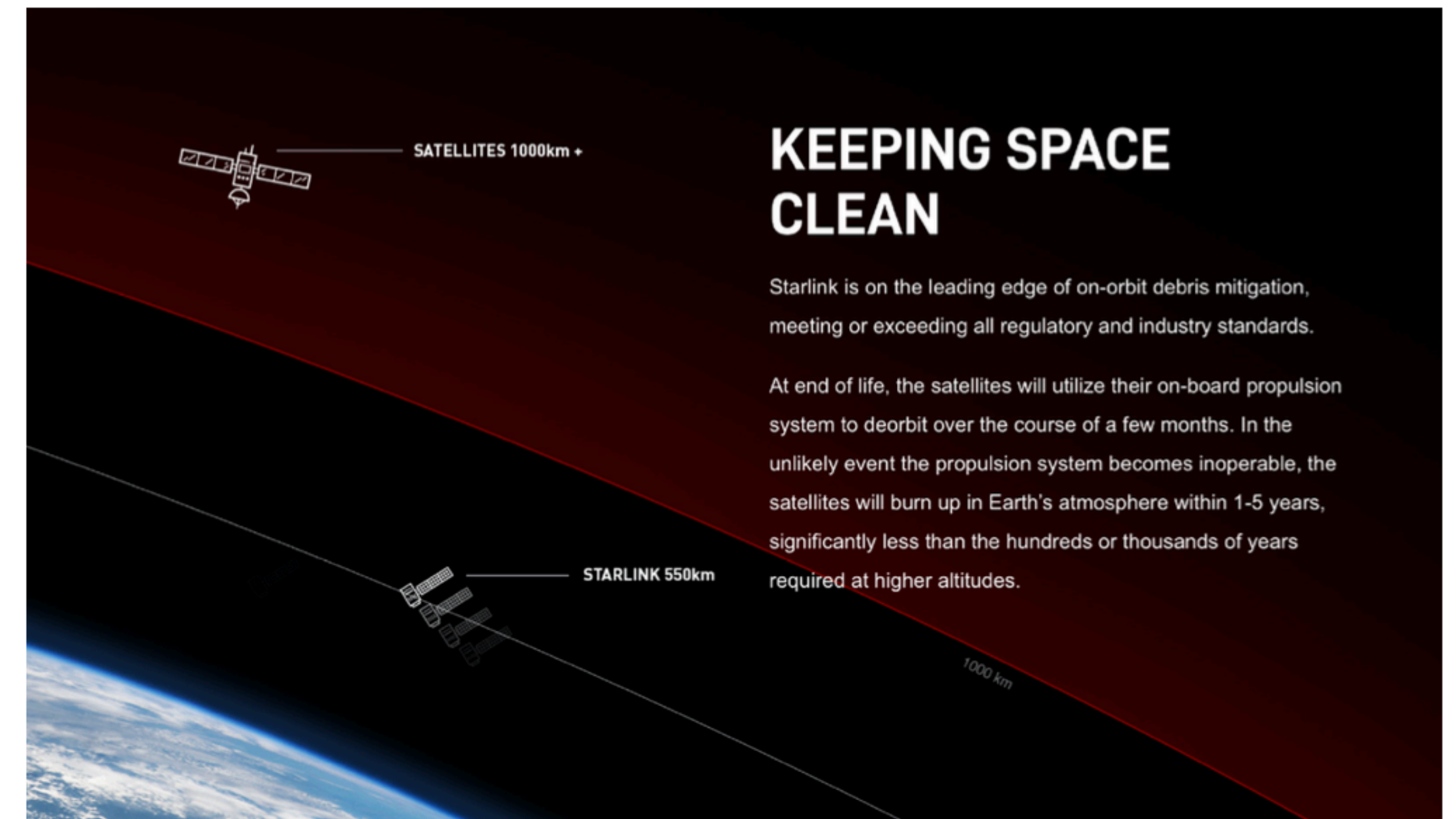


<https://engineering.fb.com/production-engineering/introducing-data-center-fabric-the-next-generation-facebook-data-center-network/>

Innovations in network design

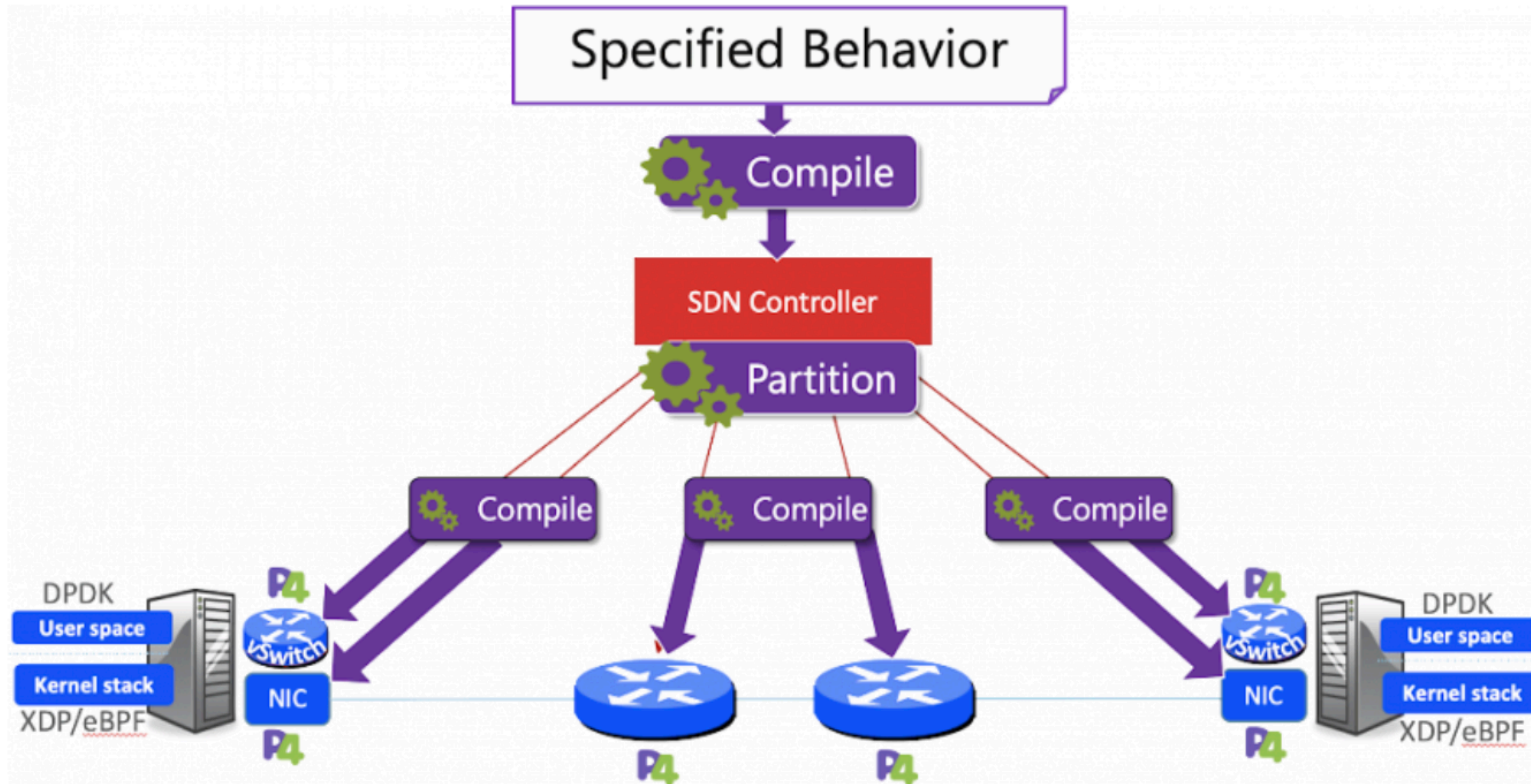


<https://loon.com> (discontinued)



<https://www.starlink.com>

Innovations in network management



<https://prontoproject.org>

When ML/RL meets networking

Imagine a network that can program itself
based on high level policies of its operator

A Self-Programming Network Will:

- do its own configuration, maintenance, and verification
- maximize its resource utilization
- provide differentiated and deterministic SLAs to applications and users
- support varying workloads
- provide an interactive dashboard for users and operators

all by itself with high level policies by the operator

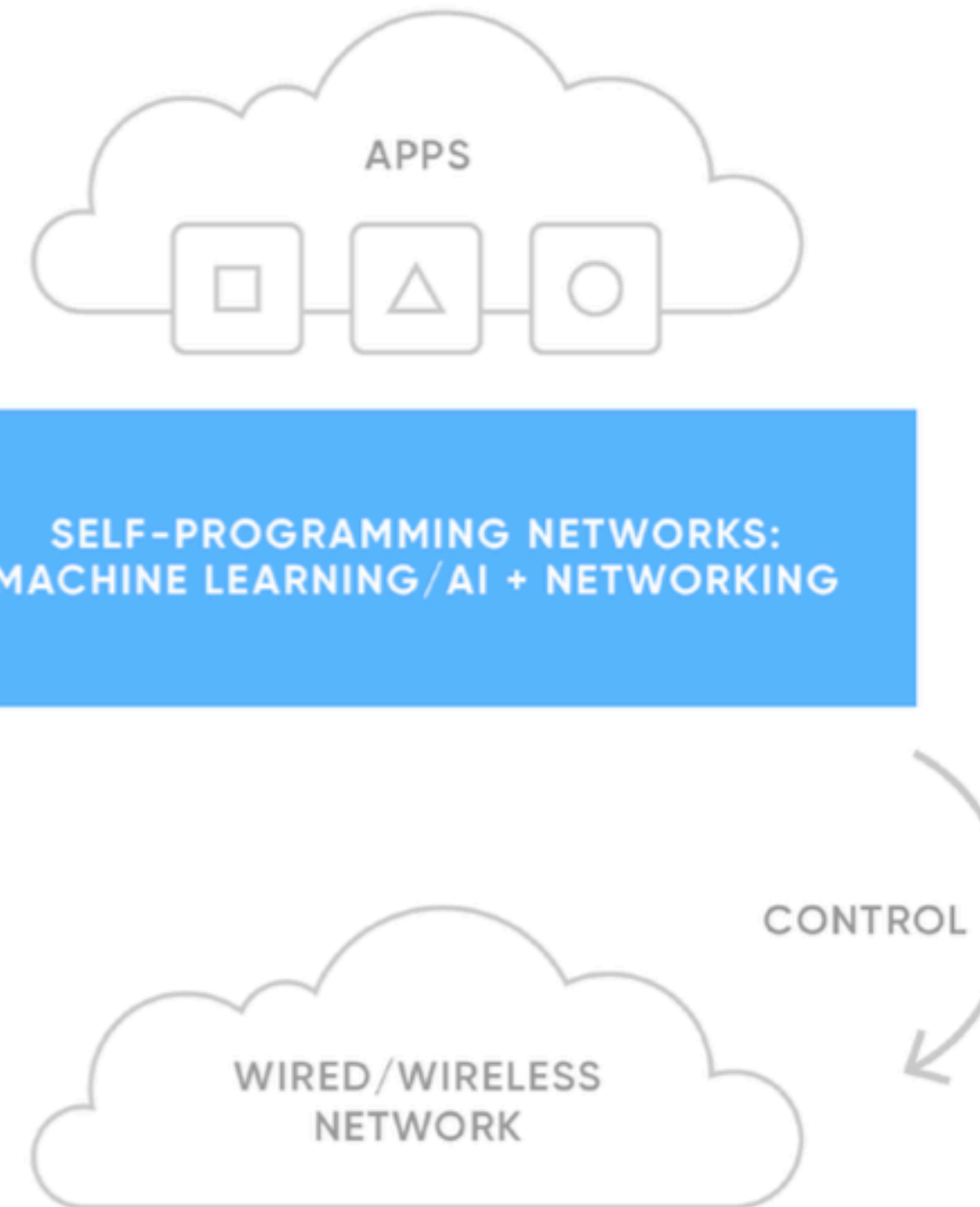
OPERATOR
POLICIES

SELF-PROGRAMMING NETWORKS:
MACHINE LEARNING/AI + NETWORKING

SENSE

CONTROL

WIRED/WIRELESS
NETWORK



Questions?